

CHASE RUBIN

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EDUCATION

University of Massachusetts Amherst

College Of Engineering

Bachelor of Science in Electrical Engineering

Master of Science in Electrical Engineering

Amherst, MA

Cumulative GPA: 3.4

Candidate, May 2027

Candidate, May 2028

- Relevant Coursework: Intro to VLSI Design Principals, Circuits and Electronics I, II, & III, Embedded systems, Synthesis and Verification of Digital Systems, Microwave Engineering I, Signal Processing Methods, Semiconductor Devices

SKILLS

Verilog, LT-Spice/H-Spice, Standard Cell Design Cadence Innovus, Synopsys Design Compiler, Embedded systems, Cadence Virtuoso, Circuit/PCB Design, Soldering, SolidWorks Electrical, Electrical Wiring, Altium Designer, STM-32, Python, C

EXPERIENCE

Axcelis Technologies | *Analog Circuit Design, Circuit Testing, Altium, LT-Spice, Soldering* *Electrical Engineering Intern (R&D)*

Beverly, MA

June 2026 – Present

- Designed an analog control system to regulate a VCXO's temperature (10-40°C) to $\pm 0.2^\circ\text{C}$ using a Peltier device
- Built a feedback circuit comparing thermocouple readback to a setpoint, generating an amplified error signal
- Used the error signal to select the direction and strength of an H-Bridge using a PWM generator and window comparators
- Validated the design in SPICE, then built and tested each stage on a breadboard and subsequently a soldered breadboard
- Simulated phase margin across all feedback op amps to verify closed loop stability before final validation
- Built the schematic in Altium, managing component libraries and footprints, then laid out the final PCB

Plains Technologies | *Circuit Design, Embedded Systems, C, Python* *Electrical Engineering Intern (R&D)*

Amherst, MA

June 2025 – August 2025

- Assisted in the design and development of the FlexRoE autonomous rover for agricultural research and commercialization
- Built an STM32 control system in C, enabling CAN bus communication with a Python interface
- Wired and assembled the electrical system, including relays, power supplies, computer systems, and electric motors
- Designed, soldered, and tested a custom proto board for a relay-based emergency stop and power control applications
- Collaborated with engineers to ensure reliable integration across electrical, software, and mechanical systems

PROJECTS

4 Bit Accumulator: Custom VLSI Layout | *Cadence Virtuoso, HSPICE, Pegasus DRC/LVS* *ECE 558 Intro to VLSI Design*

Amherst, MA

January 2026

- Designed and laid out a custom 4-bit ripple-carry accumulator in Cadence Virtuoso, built from transistor level bit slices
- Verified functionality via HSPICE simulation at the schematic and post layout levels achieving 0 DRC violations
- Determined maximum clock frequency (3.57 GHz) through experimental timing analysis of the critical carry path
- Characterized dynamic and static power across two supply voltages validating measured results against $P \propto V^2$ scaling

8 Bit CPU From Logic Gates in Verilog | *Verilog, Digital Design, Microarchitecture* *Personal Project*

Brookline, MA

August 2025

- Designed and implemented a fully functional 8-bit CPU using Verilog, built from primitive logic gates and flip-flops
- Developed ALU, registers, RAM, ROM, program counter, and control unit, with a custom instruction set architecture
- Wrote microcode and testbenches for simulation with Verilog and GTKWave, validating instruction, execution and timing
- Implemented load/store operations, arithmetic/logic instructions, branching, and jump functionality

Portable Sun Tracking Solar Panel | *KiCad, Soldering, Motor control, power systems, C* *Systems Engineering Semester Project*

Amherst, MA

February -May 2026

- Contributed to the design and development of a small portable sun tracking solar panel system for long outdoor adventures
- Developed a PCB using an ATMEGA-328 to manage battery safety, motor control, and implement a small user interface
- Used data sheets to find compatible modules to charge a LiPo battery and boost battery voltage to a required 5V and 1A
- Populated, tested, and verified PCB to meet all of our project specifications